DGMS Examination 2019 — Question 53 Detailed Solution

Junior Scientific Officer (Geology) | Engineering Geology & Soil Mechanics

QUESTION 53

The void ratio of the solid mass is defined as:

(a) Volume of voids / Total volume	**(b) Volume of voids / Volume of solids** **CORRECT ANSWER**
(c) Volume of solids / Volume of voids	(d) Volume of solids / Total volume

1. Executive Summary & Answer Verification

The correct option is **(b) Volume of voids / Volume of solids**.

In engineering geology, soil mechanics, and geotechnical engineering (codified by Karl Terzaghi):

- **Void Ratio (e):** Defined as the ratio of the **Volume of voids (V_v)** to the **Volume of solid mineral grains (V_s)**:

$$e = \frac{V_v}{V_s}$$

- **Porosity (n):** In contrast, porosity is defined as the ratio of the **Volume of voids (V_v)** to the **Total volume of the soil/rock mass (V)**:

$$n = \frac{V_v}{V} = \frac{V_v}{V_s + V_v}$$

Option (a) describes *porosity* (n), whereas option (b) provides the exact definition of *void ratio* (e).

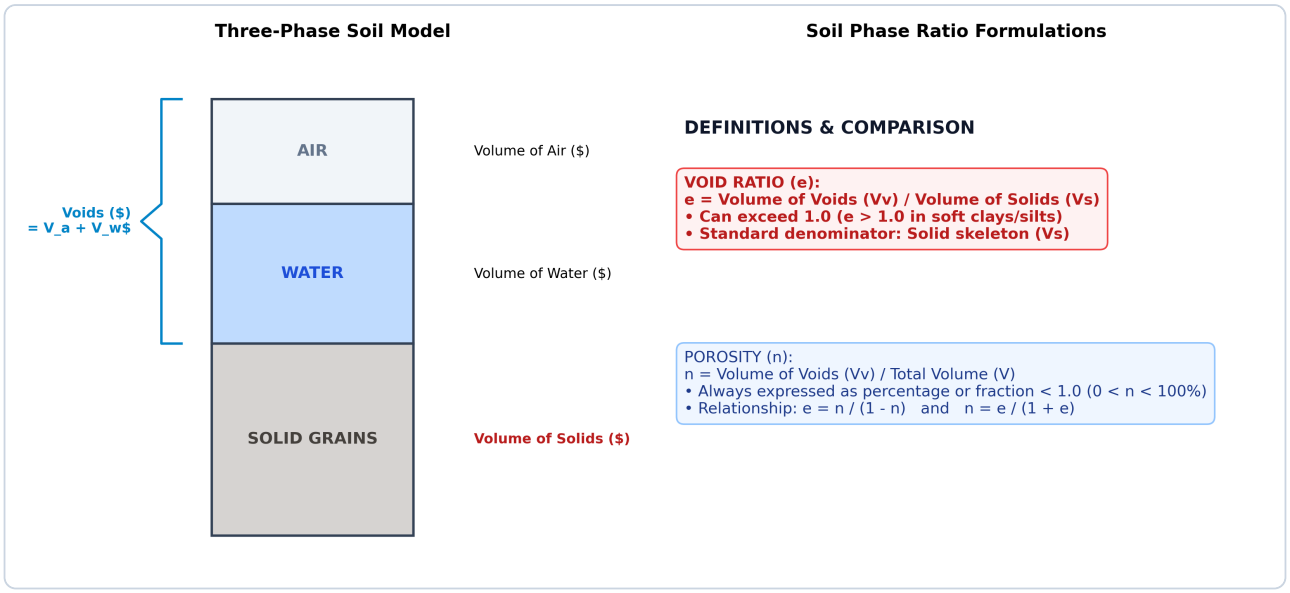


*Figure 53.1: Three-phase soil block model (Air, Water, Solid Grains) illustrating the fundamental distinction: **Void ratio ($e = V_v / V_s$)** normalizes void space against the constant solid mineral skeleton, whereas **Porosity ($n = V_v / V$)** normalizes against the variable total soil volume.*

2. Detailed Geotechnical & Mathematical Derivations

2.1 The Three-Phase Soil System

A natural particulate soil or porous rock mass consists of three distinct phases:

1. **Solid Phase (V_s, M_s):** Mineral grains (quartz, feldspar, clay minerals).
2. **Liquid Phase (V_w, M_w):** Pore water filling interstitial spaces.
3. **Gas Phase (V_a, $M_a \approx 0$):** Air filling unsaturated pores.

The total void volume (V_v) is the sum of water and air volumes:

$$V_v = V_w + V_a \quad \text{and} \quad V = V_s + V_v = V_s + V_w + V_a$$

2.2 Mathematical Interrelationship Between e and n

To derive porosity in terms of void ratio:

$$n = \frac{V_v}{V} = \frac{V_v}{V_s + V_v}$$

Dividing numerator and denominator by V_s:

$$n = \frac{\frac{V_v}{V_s}}{1 + \frac{V_v}{V_s}} = \frac{e}{1 + e}$$

Conversely, to express void ratio in terms of porosity:

$$e = \frac{n}{1 - n}$$

2.3 Why Civil & Mining Engineers Prefer Void Ratio (e) Over Porosity (n)

In consolidation, settlement, and compaction analysis:

- When a soil layer compresses under structural or overburden load, water and air are squeezed out, causing the total volume (V) to shrink.
- However, the mineral grains themselves are practically incompressible—meaning **V_s remains constant** throughout deformation.
- Because the denominator V_s in $e = V_v / V_s$ is constant, any change in void ratio (Δe) is directly proportional to the change in specimen volume (ΔV):

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\frac{\Delta V}{V_0} = \frac{\Delta e}{1 + e_0} = \frac{\Delta H}{H_0}
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- Porosity ($n = V_v / V$) has a variable denominator, making settlement mathematics non-linear and cumbersome.

3. Comprehensive Comparison Table of Phase Parameters

Property	Symbol	Mathematical Formula	Typical Range	Physical Meaning
Void Ratio	**e**	**V_v / V_s**	0.3 – 1.5 (can exceed 2.0+ in clays/peat)	**Volume of voids per unit volume of solids (Option b)**
Porosity	n	V_v / V	20% – 60% (always $0 < n < 1.0$)	Volume of voids per unit total volume (Option a)
Degree of Saturation	S_r	V_w / V_v	0% (dry) to 100% (fully saturated)	Fraction of void space occupied by water
Air Content	a_c	V_a / V_v	0% to 100%	Fraction of void space occupied by air ($a_c = 1 - S_r$)

4. Evaluation of All Multiple Choice Options

- **(a) Volume of voids / Total volume — INCORRECT:** This is the definition of **Porosity** (n), NOT void ratio.
- **(b) Volume of voids / Volume of solids — CORRECT:** This is the universal, standardized definition of **Void Ratio** ($e = V_v / V_s$).
- **(c) Volume of solids / Volume of voids — INCORRECT:** This is the inverse of void ratio ($1/e$).
- **(d) Volume of solids / Total volume — INCORRECT:** Equal to $1 - n$, representing the solid volume fraction.

Textbook Authority

Karl Terzaghi & Ralph B. Peck, *Soil Mechanics in Engineering Practice*:
"The void ratio e is defined as the ratio of the volume of the voids to the volume of the solid substance. The porosity n is the ratio of the volume of the voids to the total volume."
K. M. Bangar, *Principles of Engineering Geology*:

"Void ratio (e) = Volume of voids / Volume of solids = Vv / Vs. Porosity (n) = Volume of voids / Total volume = Vv / V."